Problem 2: MARS ROVER SAMPLE COLLECTION PLANNING USING PDDL

Task 3: Plan Generation Metrics

The following plans were generated using the Fast Downward planning system. The Satisficing configurations utilized Greedy Best-First Search with the FF heuristic, while the Optimal configurations utilized A* Search with the Landmark-Cut (LM-cut) heuristic.

Scenario	Configuration	Plan Length	States Expanded	Planning Time (s)
Basic	Satisficing (Greedy)	30	47	0.0025
Basic	Optimal (A*)	25	5,776	0.6267
Extended	Satisficing (Greedy)	46	91	0.0048
Extended	Optimal (A*)	40	2,230,060	347.9179

**Note: Total number of actions equals plan length in all generated sequential plans.*

Task 6: Plan Analysis

1. Comparison of Satisficing and Optimal Plan Lengths

> The satisficing search produces strictly longer plans (30 steps basic / 46 steps extended) compared to the optimal A* search (25 steps basic / 40 steps extended). The satisficing planner chases immediate heuristic rewards, resulting in inefficient behaviors like picking up a single rock, driving all the way back to the base to deposit it, and returning with an empty second bay. The optimal planner, however, computes the global minimum cost. It successfully batches collections, ensuring the rover stays out in the field until both bays are full before undertaking the costly return trip to the base.

2. Optimal Use of Bays and the Drill

> The optimal plan maximizes carrying capacity by simultaneously utilizing both `Bay1` and `Bay2`, reducing total travel distance. Crucially, the optimal planner forces the rover to strategically detour to `SiteE` to execute the `pick_drill` action early in the mission (often combined with a trip to collect soft rocks in the same vicinity). By acquiring the drill before attempting to clear sites containing hard rocks, the rover avoids wasted round trips and back-tracking.

3. Extended Scenario: Dust Storm Routing

> In the extended scenario, the dust storm eliminates the direct two-way `Base ↔ SiteB` route. To reach `SiteB` and `SiteD`, the optimal plan routes the rover through `Base → SiteA → SiteC`, and then explicitly utilizes the steep one-way slope (`SiteC → SiteD`). After collecting rocks at `SiteB` and `SiteD`, the rover is unable to climb back up the steep slope to return. Instead, it must navigate forward to exit the cluster via `SiteE`, looping back through `SiteC → SiteA → Base`.